

Tutorial for Chapter 3

Exercise 1.

For each of the following, determine whether the given values can serve as the values of a probability distribution of a random variable with the range $x = 1, 2, 3$ and 4 :

1. $f(1) = 0.25, f(2) = 0.75, f(3) = 0.25, f(4) = -0.25$.
2. $f(1) = 0.15, f(2) = 0.27, f(3) = 0.29, f(4) = 0.29$.
3. $f(1) = \frac{1}{19}, f(2) = \frac{10}{19}, f(3) = \frac{2}{19}, f(4) = \frac{5}{19}$.

Solution 1.

1. No, because $f(4)$ is negative.
2. Yes, as $f(1) + f(2) + f(3) + f(4) = 0.15 + 0.27 + 0.29 + 0.29 = 1$.
3. No, as $f(1) + f(2) + f(3) + f(4) = \frac{1+10+2+5}{19} = \frac{18}{19}$.

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Exercise 2.

Let X have the following PMF:

$$P(X = x) = \begin{cases} 0.1 & x = 1 \\ 0.3 & x = 2 \\ 0.4 & x = 3 \\ 0.2 & x = 4 \\ 0 & \text{otherwise} \end{cases}$$

1. Verify that this is a valid PMF.
2. Find $P(X \leq 2)$.
3. Find $P(2 < X \leq 4)$.

Solution 2.

1. Sum of probabilities:

$$0.1 + 0.3 + 0.4 + 0.2 = 1.0 \Rightarrow \text{Valid PMF}$$

2. $P(X \leq 2) = P(1) + P(2) = 0.1 + 0.3 = 0.4$
3. $P(2 < X \leq 4) = P(3) + P(4) = 0.4 + 0.2 = 0.6$

Exercise 3.

Let X be the outcome when a fair six-sided die is rolled.

- Write the PMF of X .
- Find $P(X > 4)$.

Solution 3.

- PMF: $P(X = x) = \frac{1}{6}$ for $x = 1, 2, 3, 4, 5, 6$
- $P(X > 4) = P(5) + P(6) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

Exercise 4.

A coin is biased so that heads is twice as likely as tails. For three independent tosses of the coin, find

1. the probability distribution of X , the total number of heads;
2. the probability of getting at most two heads.

Solution 4.

We have $P(H) = \frac{2}{3}$, and the range of X is $\{0, 1, 2, 3\}$.

1. The probability distribution of X is given by

- $P(X = 0) = P(TTT) = P(T)P(T)P(T) = \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$
- $P(X = 1) = 3 \times P(H)P(T)P(T) = 3 \times \frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{6}{27}$
- $P(X = 2) = 3 \times P(H)P(H)P(T) = 3 \times \frac{2}{3} \times \frac{2}{3} \times \frac{1}{3} = \frac{12}{27}$
- $P(X = 3) = P(HHH) = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$

2. $P(X \geq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{1}{27} + \frac{6}{27} + \frac{12}{27} = \frac{19}{27}$

Exercise 5.

Construct a probability histogram for the following probability distributions,

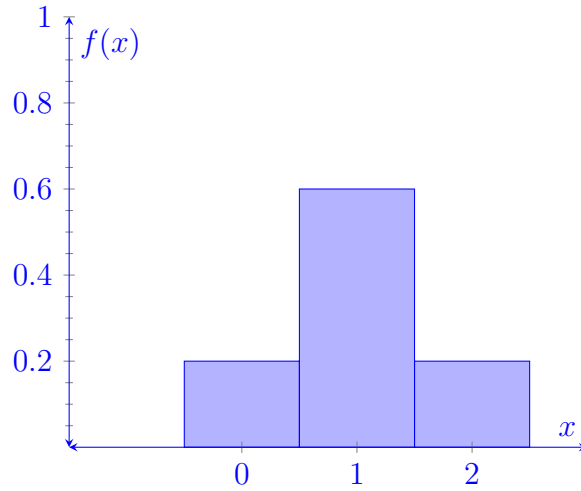
$$f(x) = \frac{\binom{2}{x} \binom{4}{3-x}}{\binom{6}{3}}, \quad x = 0, 1, 2.$$

Solution 5.

The probability for each value of x is

$$f(0) = \frac{\binom{2}{0} \binom{4}{3}}{\binom{6}{3}} = \frac{4}{20} = \frac{1}{5}, \quad f(1) = \frac{\binom{2}{1} \binom{4}{2}}{\binom{6}{3}} = \frac{12}{20} = \frac{3}{5}, \quad f(2) = \frac{\binom{2}{2} \binom{4}{1}}{\binom{6}{3}} = \frac{4}{20} = \frac{1}{5}.$$

Below is the histogram,



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Exercise 6.

Let X be a discrete random variable with:

$$P(X = x) = kx, \quad x = 1, 2, 3, 4$$

1. Find the value of k .
2. Find $P(X \geq 3)$.

Solution 6.

1. We require: $\sum_{x=1}^4 kx = 1 \Rightarrow k(1 + 2 + 3 + 4) = 1 \Rightarrow k \cdot 10 = 1 \Rightarrow k = \frac{1}{10}$
2. $P(X \geq 3) = P(3) + P(4) = \frac{1}{10}(3 + 4) = \frac{7}{10}$

Exercise 7.

Let X with the following PMF:

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

1. Prove that this is a valid PMF.
2. For $p = \frac{1}{3}$, compute $P(X > 3)$.

Solution 7.

1. Proof:

$$\sum_{k=1}^{\infty} (1 - p)^{k-1}p = p \sum_{k=0}^{\infty} (1 - p)^k = \frac{p}{1 - (1 - p)} = 1$$

2. For $p = \frac{1}{3}$:

$$P(X > 3) = (1 - p)^3 = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

Exercise 8.

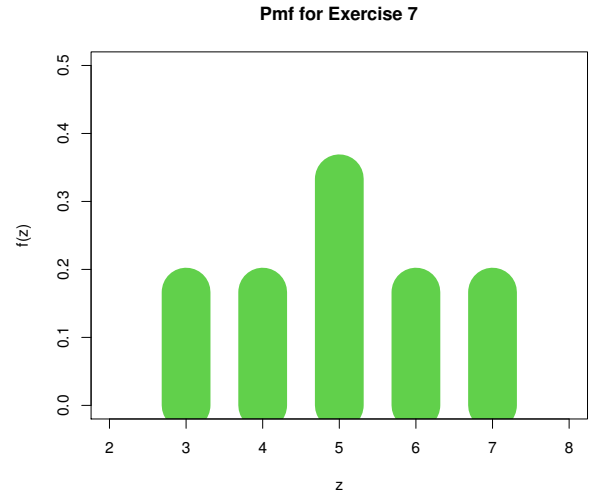
An urn contains four balls numbered 1, 2, 3, and 4. If two balls are drawn from the urn at random (that is, each pair has the same chance of being selected) and Z is the sum of the numbers on the two balls drawn, find

1. the probability distribution of Z and draw a histogram;
2. the distribution function of Z and draw its graph.

Solution 8.

1. The probability distribution of Z is summarized below

Draw	B_1	B_2	z	$P(Z = z)$
1	1	2	3	2/12
2	1	3	4	2/12
3	1	4	5	4/12
4	2	1	3	2/12
5	2	3	5	4/12
6	2	4	6	2/12
7	3	1	4	2/12
8	3	2	5	4/12
9	3	4	7	2/12
10	4	1	5	4/12
11	4	2	6	2/12
12	4	3	7	2/12



2. For the cumulative distribution function, we have

$$F(z) = \begin{cases} 0 & \text{if } z < 3 \\ \frac{2}{12} & \text{if } 3 \leq z < 4 \\ \frac{4}{12} & \text{if } 4 \leq z < 5 \\ \frac{8}{12} & \text{if } 5 \leq z < 6 \\ \frac{10}{12} & \text{if } 6 \leq z < 7 \\ 1 & \text{if } z \geq 7 \end{cases}$$

Exercise 9.

Find the distribution function of the random variable that has the probability distribution

$$f(x) = \frac{x}{15}, \quad x = 1, 2, 3, 4, 5.$$

Solution 9.

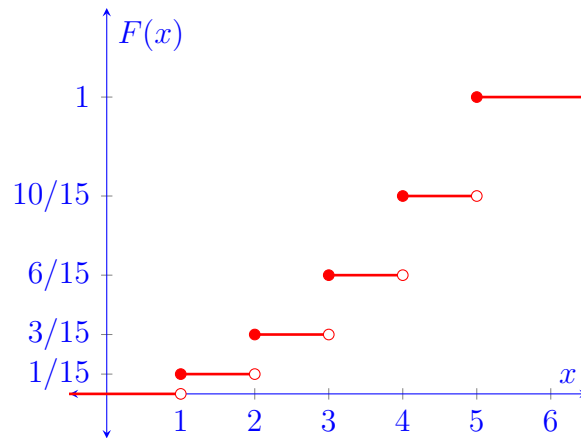
Firstly, we write down all the pmf explicitly,

$$f(1) = \frac{1}{15}, \quad f(2) = \frac{2}{15}, \quad f(3) = \frac{3}{15}, \quad f(4) = \frac{4}{15}, \quad f(5) = \frac{5}{15}.$$

The distribution function is then as follows,

$$F(x) = \begin{cases} 0, & x < 1, \\ 1/15, & 1 \leq x < 2, \\ 3/15, & 2 \leq x < 3, \\ 6/15, & 3 \leq x < 4, \\ 10/15, & 4 \leq x < 5, \\ 1, & 5 \leq x. \end{cases}$$

Below is the graph of cdf,



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Exercise 10.

If X has the cumulative distribution function

$$F(x) = \begin{cases} 0, & x < 1, \\ 1/3, & 1 \leq x < 4, \\ 1/2, & 4 \leq x < 6, \\ 5/6, & 6 \leq x < 10, \\ 1, & x \geq 10. \end{cases}$$

Find

1. $P(2 < X \leq 6)$;
2. $P(X = 4)$;
3. the probability distribution of X .

Solution 10.

1.

$$P(2 < X \leq 6) = F(6) - F(1) = \frac{5}{6} - \frac{1}{3} = \frac{1}{2}.$$

2.

$$P(X = 4) = F(4) - F(1) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

3. One has

$$P(X = 1) = F(1) = \frac{1}{3}, \quad P(X = 6) = F(6) - F(4) = \frac{5}{6} - \frac{1}{2} = \frac{1}{3},$$

and

$$P(X = 10) = F(10) - F(6) = 1 - \frac{5}{6} = \frac{1}{6}.$$

The pmf is

$$f(x) = \begin{cases} 1/3, & x = 1, \\ 1/6, & x = 4, \\ 1/3, & x = 6, \\ 1/6, & x = 10. \end{cases}$$

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